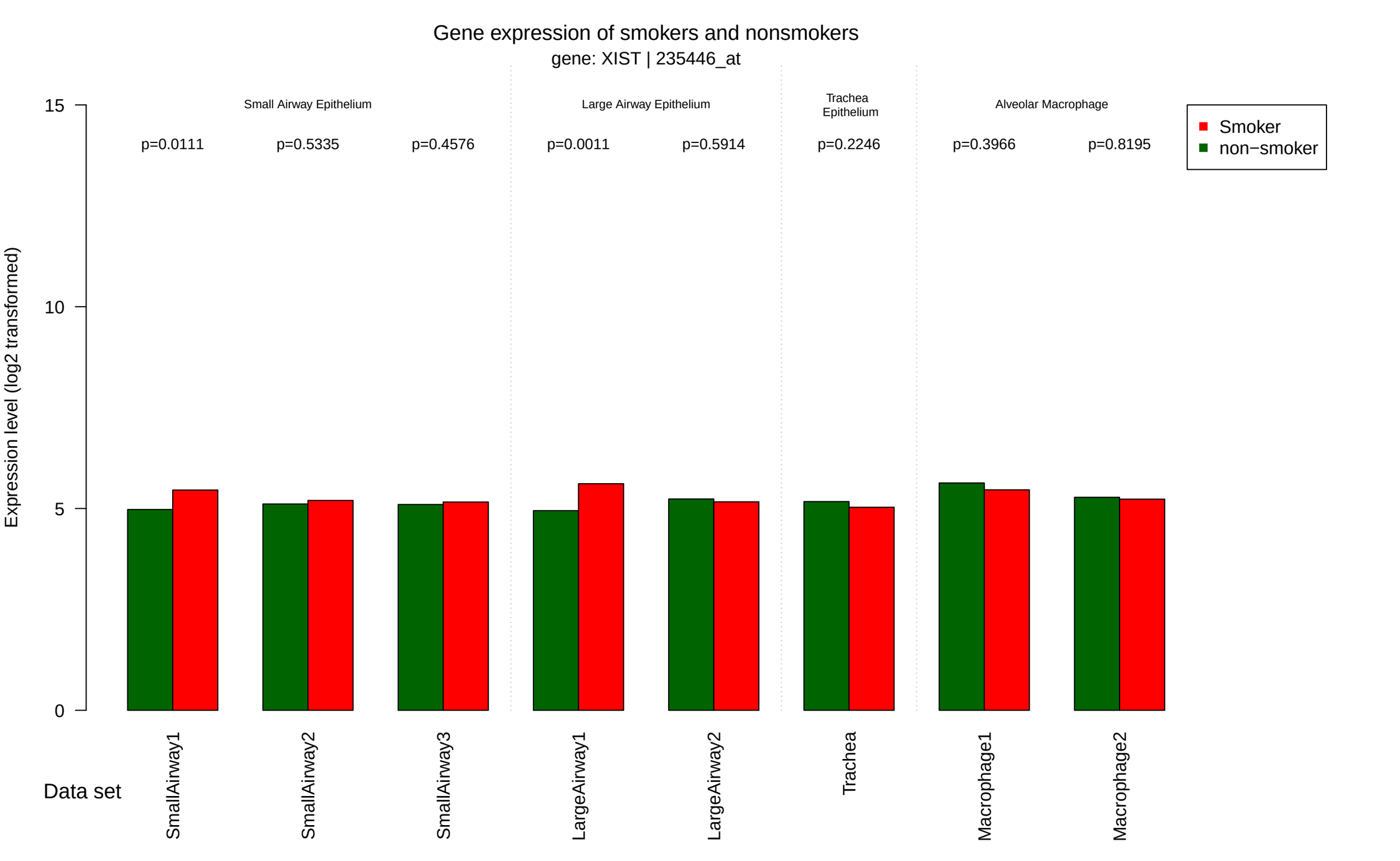
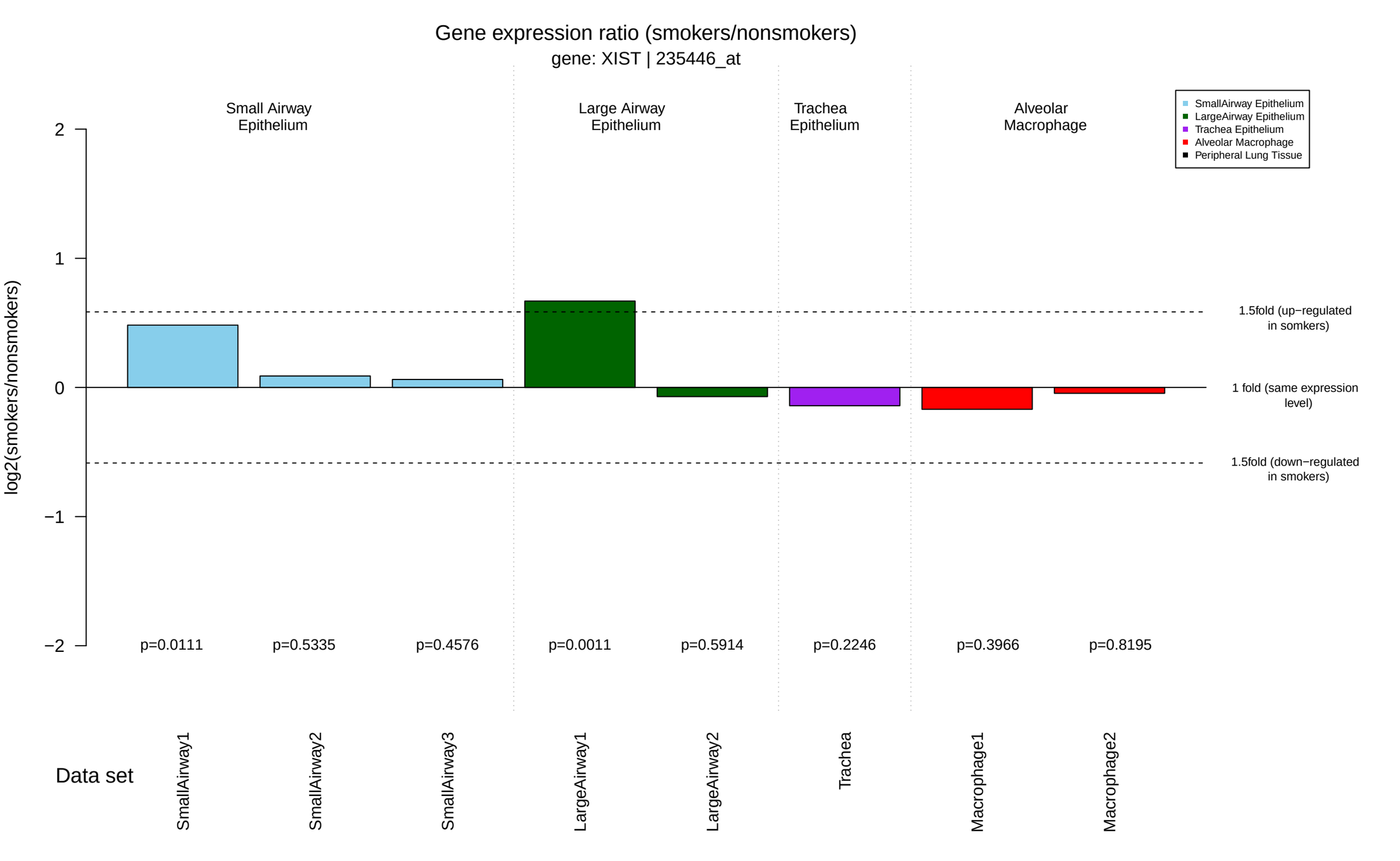
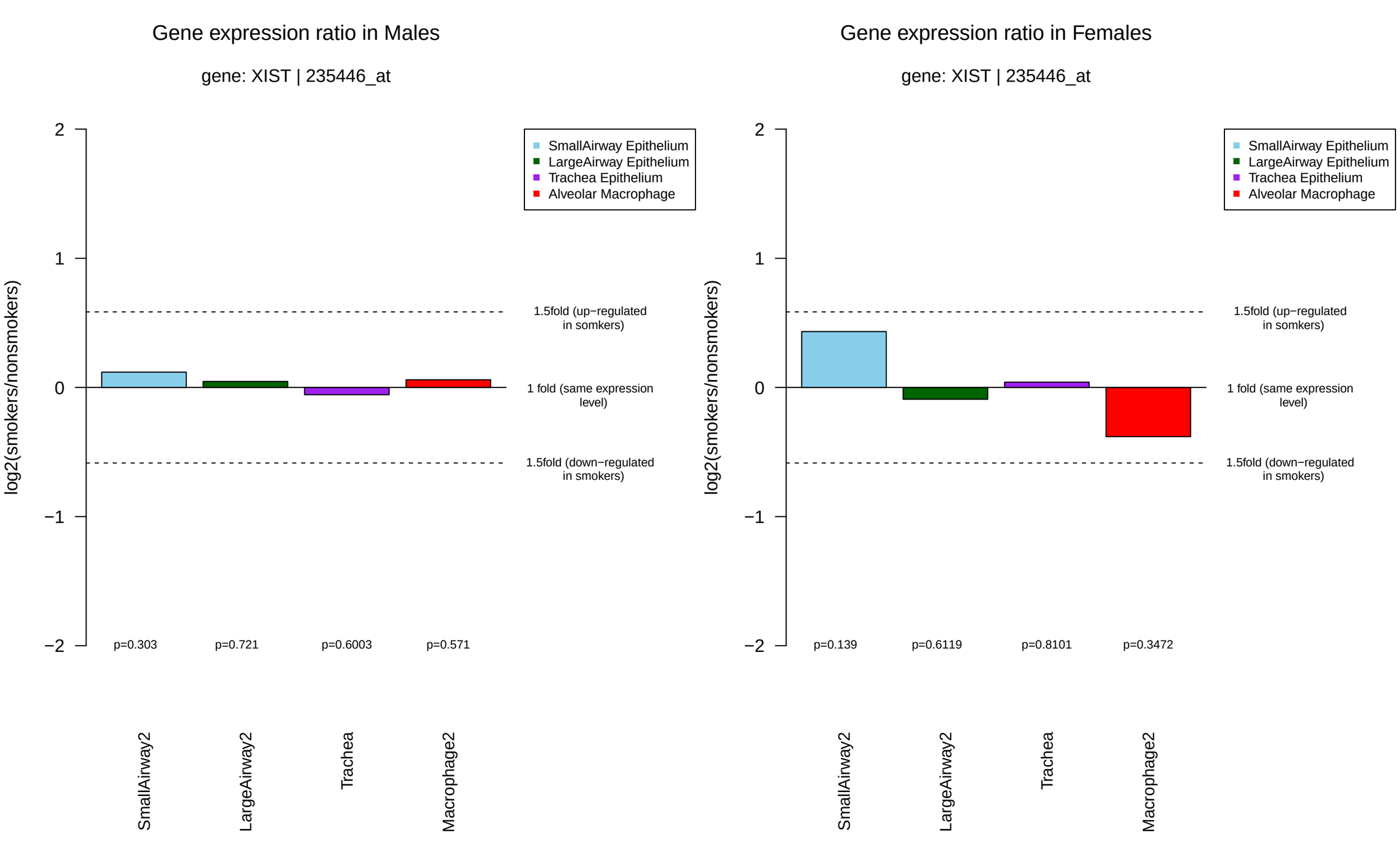
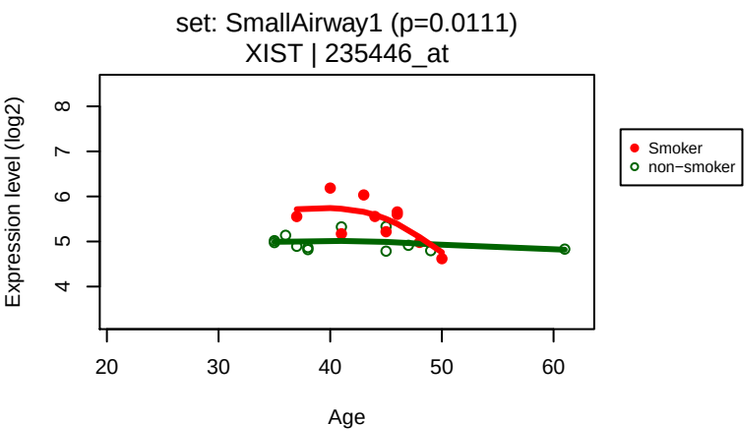
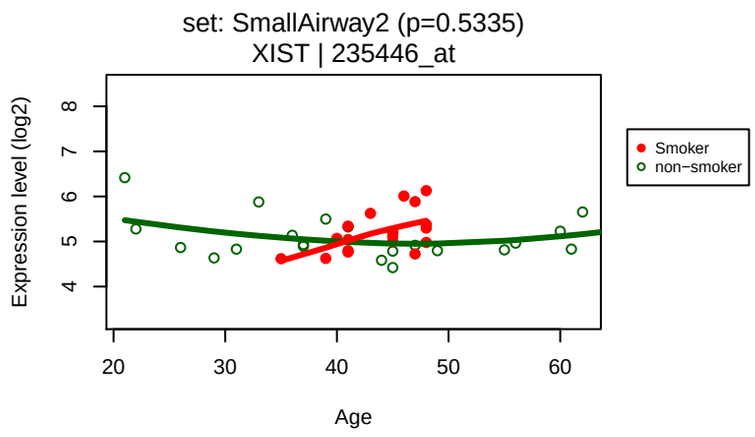
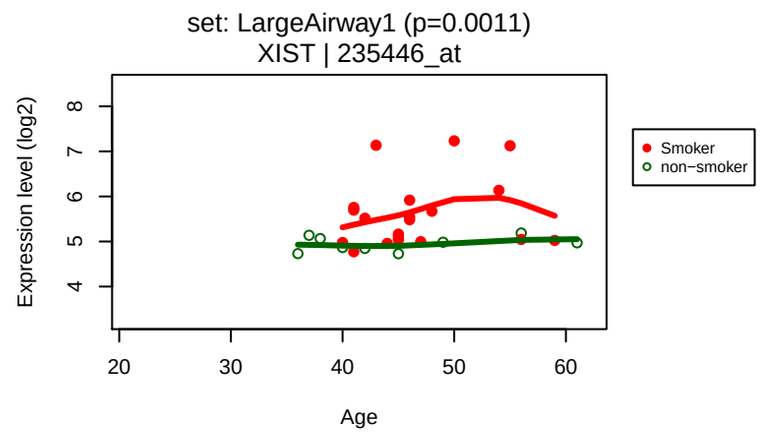

Gene expression of smokers and nonsmokers
gene: XIST | 235446_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0111
p=0.5335
p=0.4576
p=0.0011
p=0.5914
p=0.2246
p=0.3966
p=0.8195
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

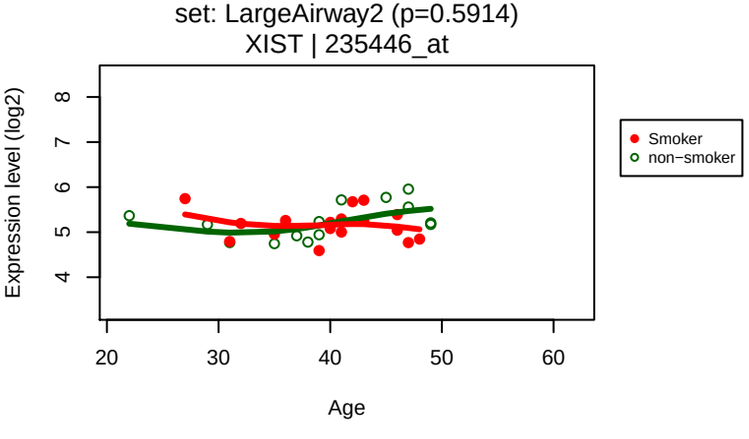

Gene expression ratio (smokers/nonsmokers)
gene: XIST | 235446_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0111
p=0.5335
p=0.4576
p=0.0011
p=0.5914
p=0.2246
p=0.3966
p=0.8195
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

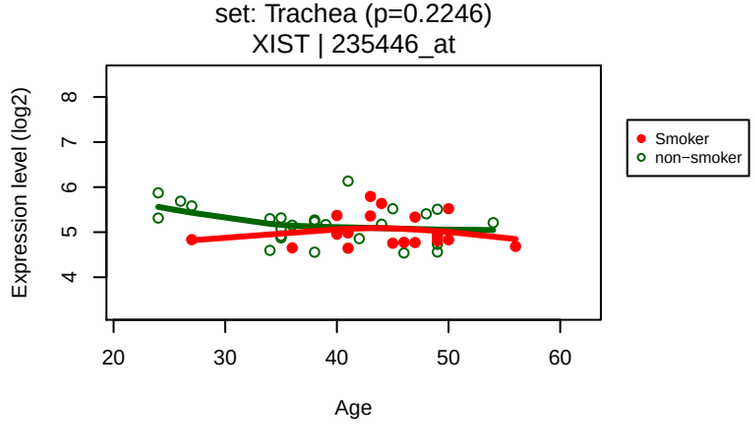

Gene expression ratio in Males
gene: XIST | 235446_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.303
p=0.721
p=0.6003
p=0.571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XIST | 235446_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.139
p=0.6119
p=0.8101
p=0.3472
SmallAirway2
LargeAirway2
Trachea
Macrophage2

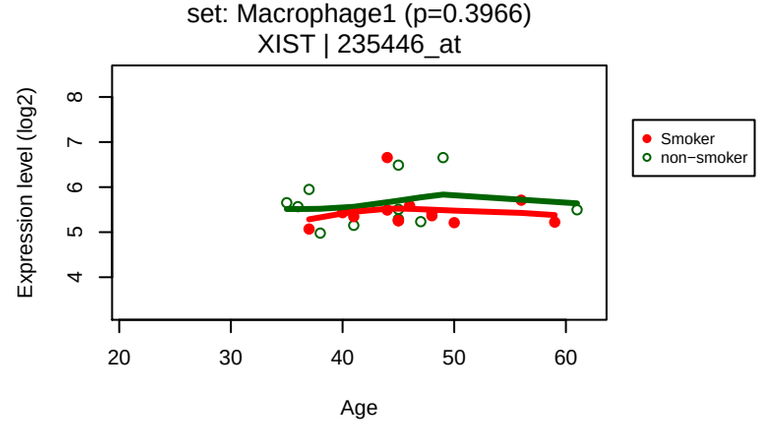

set: SmallAirway1 (p=0.0111)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker

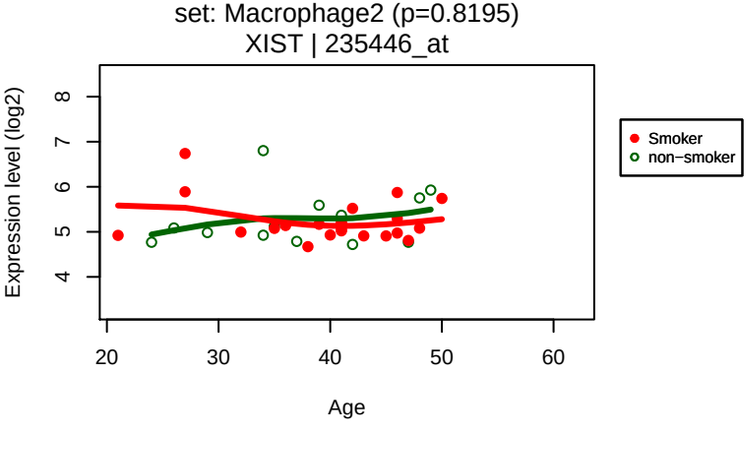

set: SmallAirway2 (p=0.5335)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker

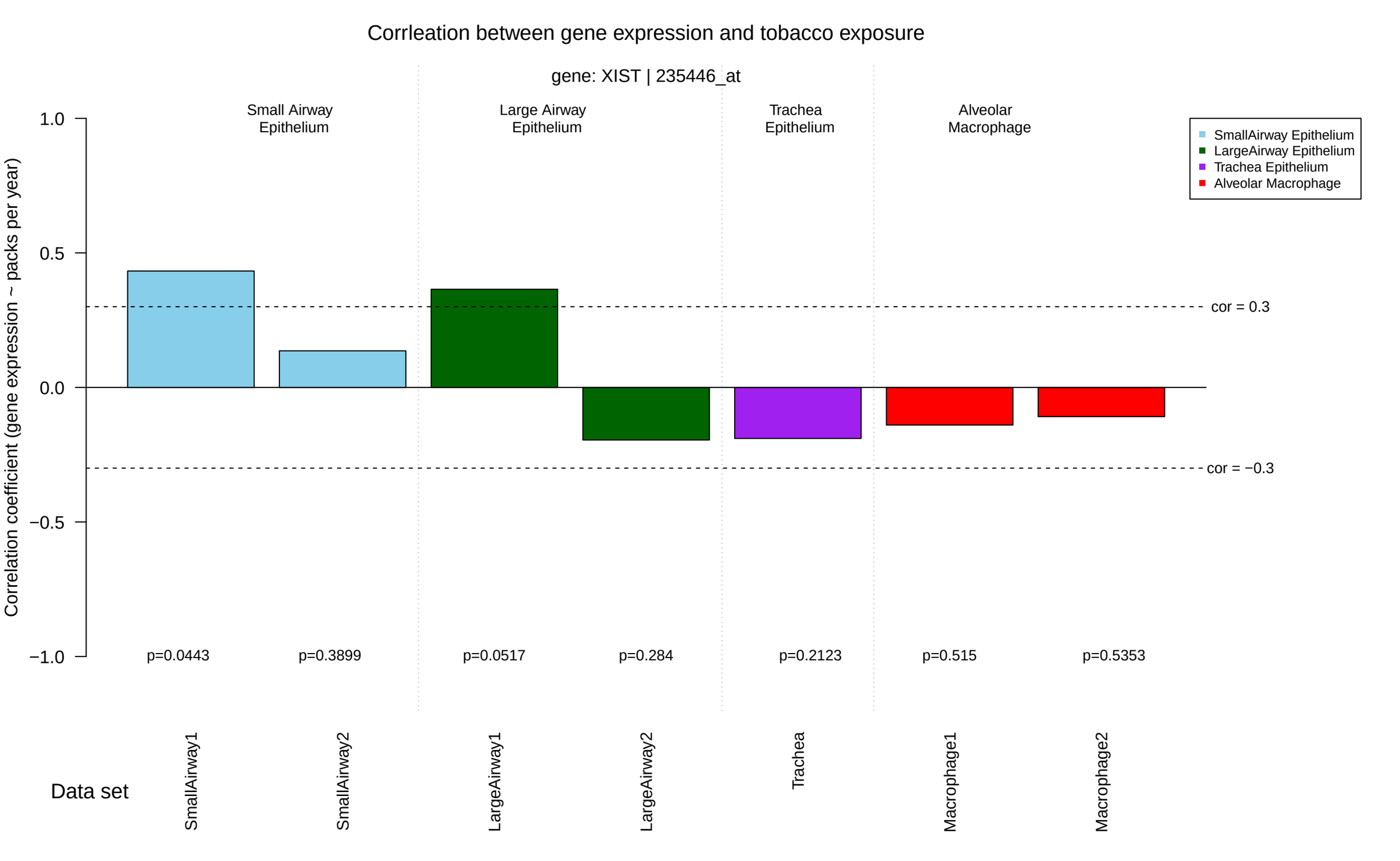

set: LargeAirway1 (p=0.0011)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5914)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2246)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3966)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8195)
XIST | 235446_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XIST | 235446_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0443
p=0.3899
p=0.0517
p=0.284
p=0.2123
p=0.515
p=0.5353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2